

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [3]: df = pd.read_csv("train.csv")
df.head()
```

Out[3]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500



```
In [ ]: The dataset was loaded successfully. The first five rows display passenger details
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In []:

The Titanic dataset was loaded successfully. The first five rows display passenger Survived, Pclass, Name, Sex, Age, Fare, and Embarked. This provides an initial understanding of available features.

In [4]:

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    str
4   Sex          891 non-null    str
5   Age         714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    str
9   Fare         891 non-null    float64
10  Cabin        204 non-null    str
11  Embarked     889 non-null    str
dtypes: float64(2), int64(5), str(5)
memory usage: 83.7 KB
```

In []: The dataset contains passenger information such as age, gender, class, fare, and survival. Some columns have missing values

In [5]: `df.describe()`

Out[5]:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204200
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

In []: The statistical summary shows the mean, minimum, maximum, and distribution of numerical variables.

In [14]: `df['Survived'].value_counts()`

Out[14]:

```
Survived
0      549
1      342
Name: count, dtype: int64
```

In []: More passengers did not survive than survived.

In [7]: `df.isnull().sum()`

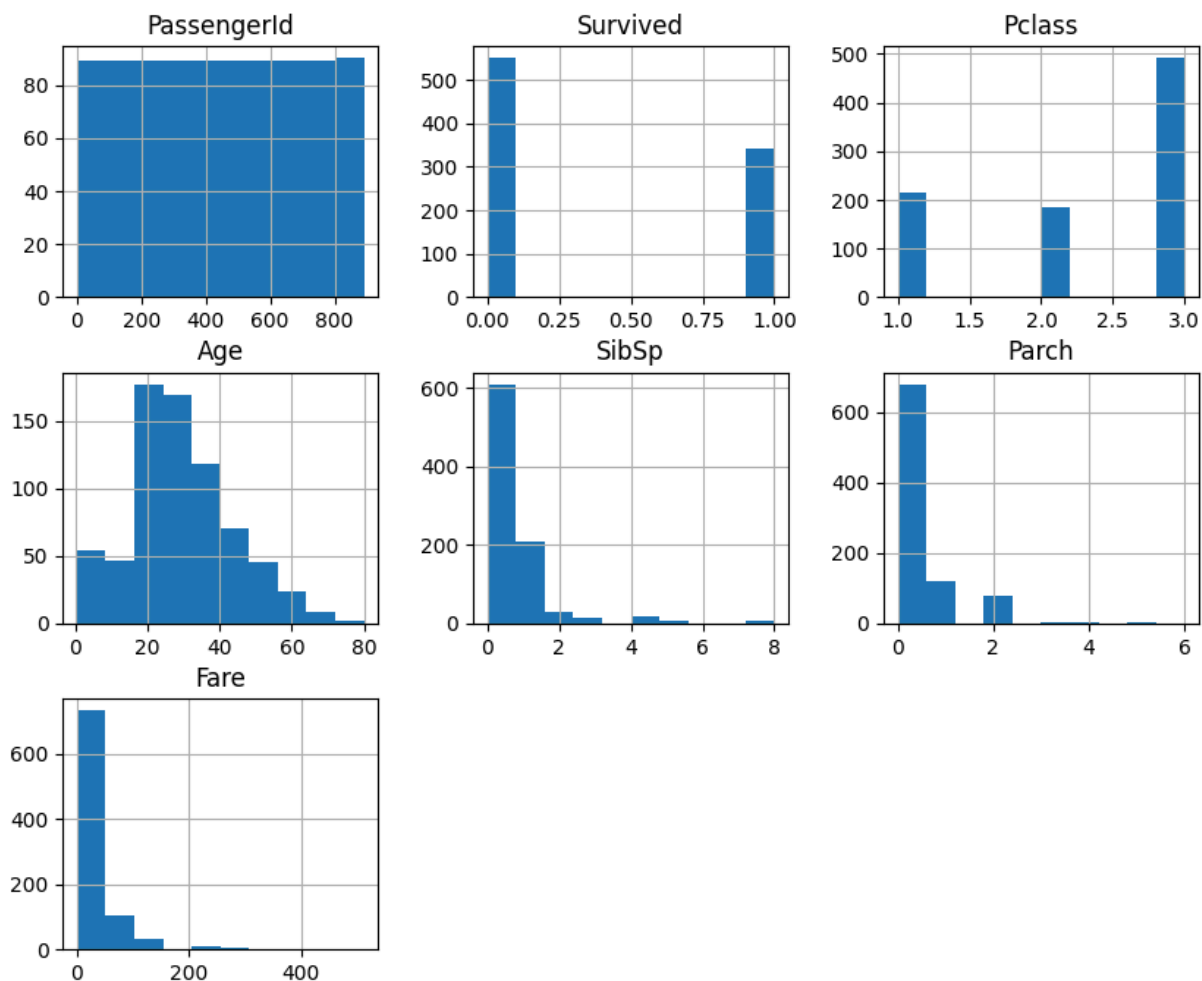
Out[7]:

```
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age          177
SibSp          0
Parch          0
Ticket         0
Fare           0
Cabin        687
Embarked       2
dtype: int64
```

In []: Missing values are present in Age, Cabin, and Embarked columns.

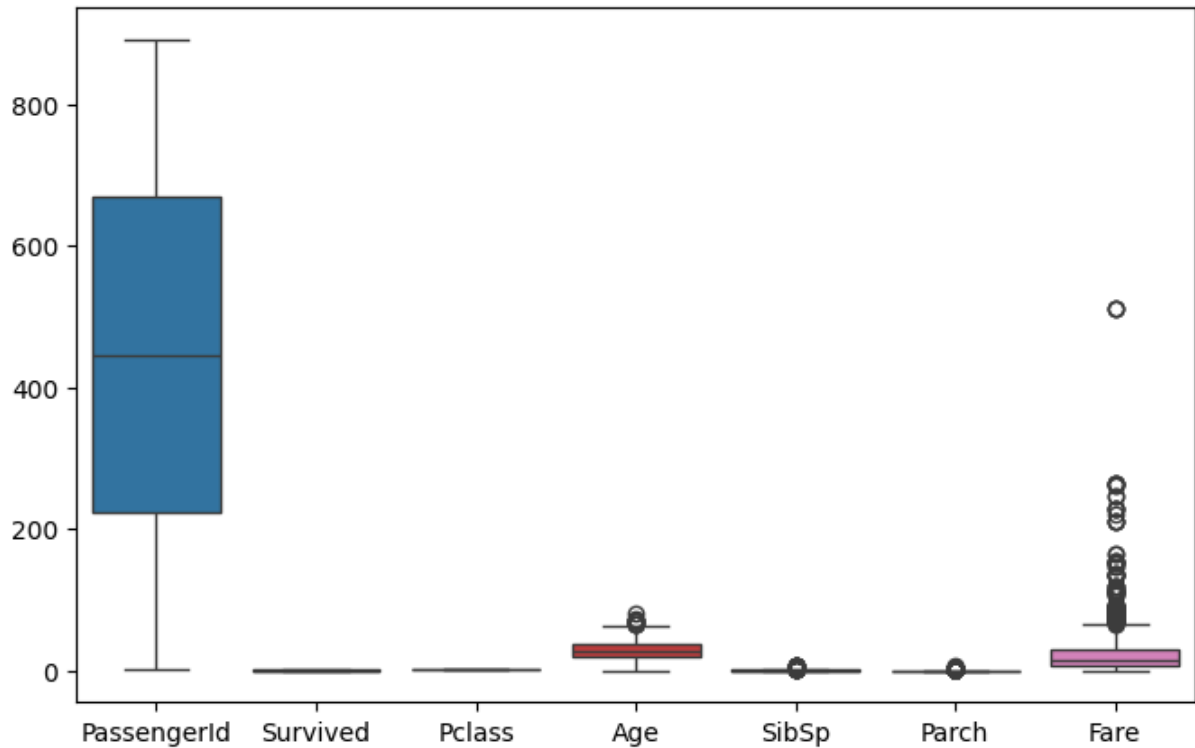
In [8]: `df.hist(figsize=(10,8))`

```
plt.show()
```



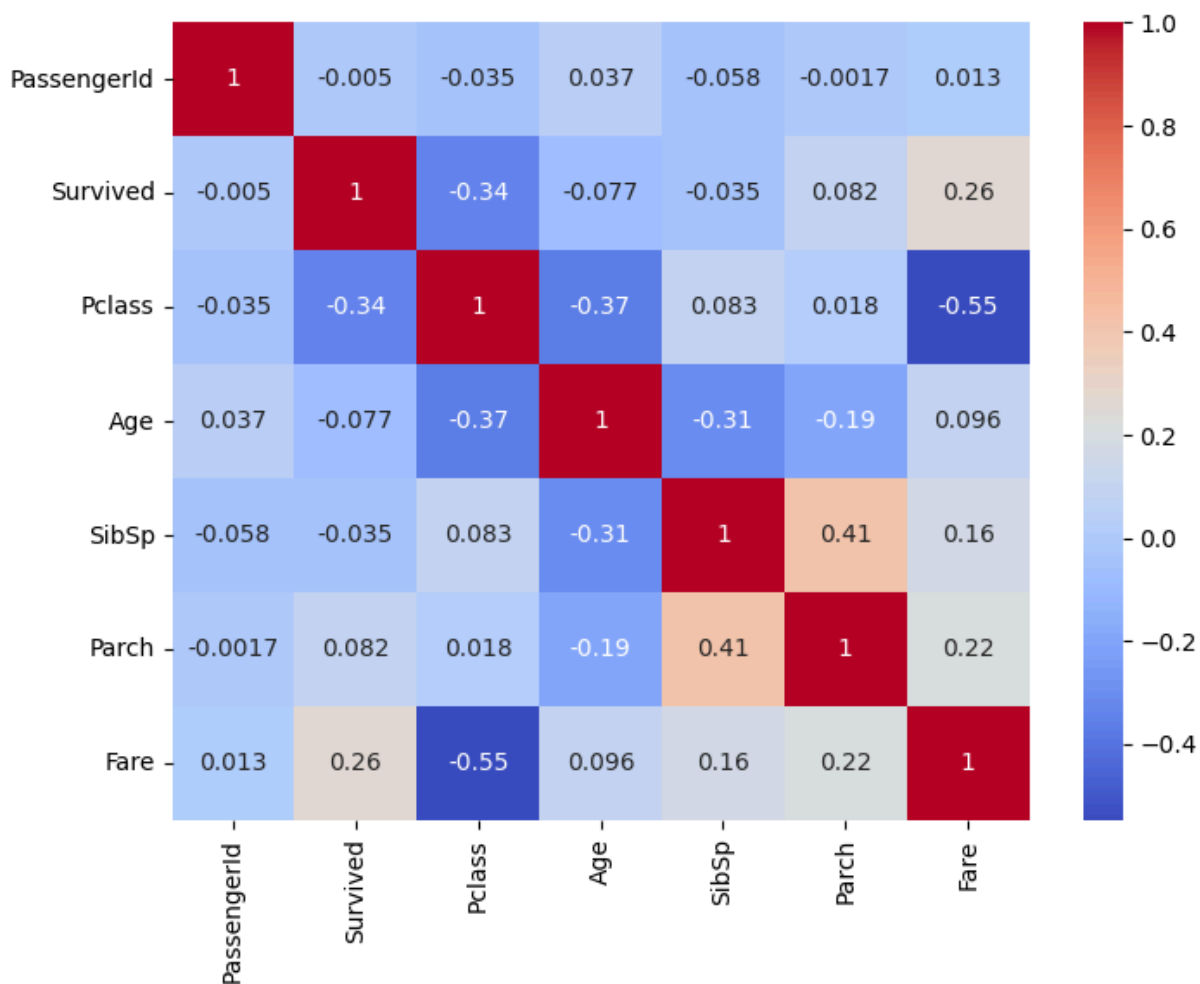
In []: The histogram shows the distribution of numerical variables such as Age and Fare.

```
In [9]: plt.figure(figsize=(8,5))
sns.boxplot(data=df)
plt.show()
```



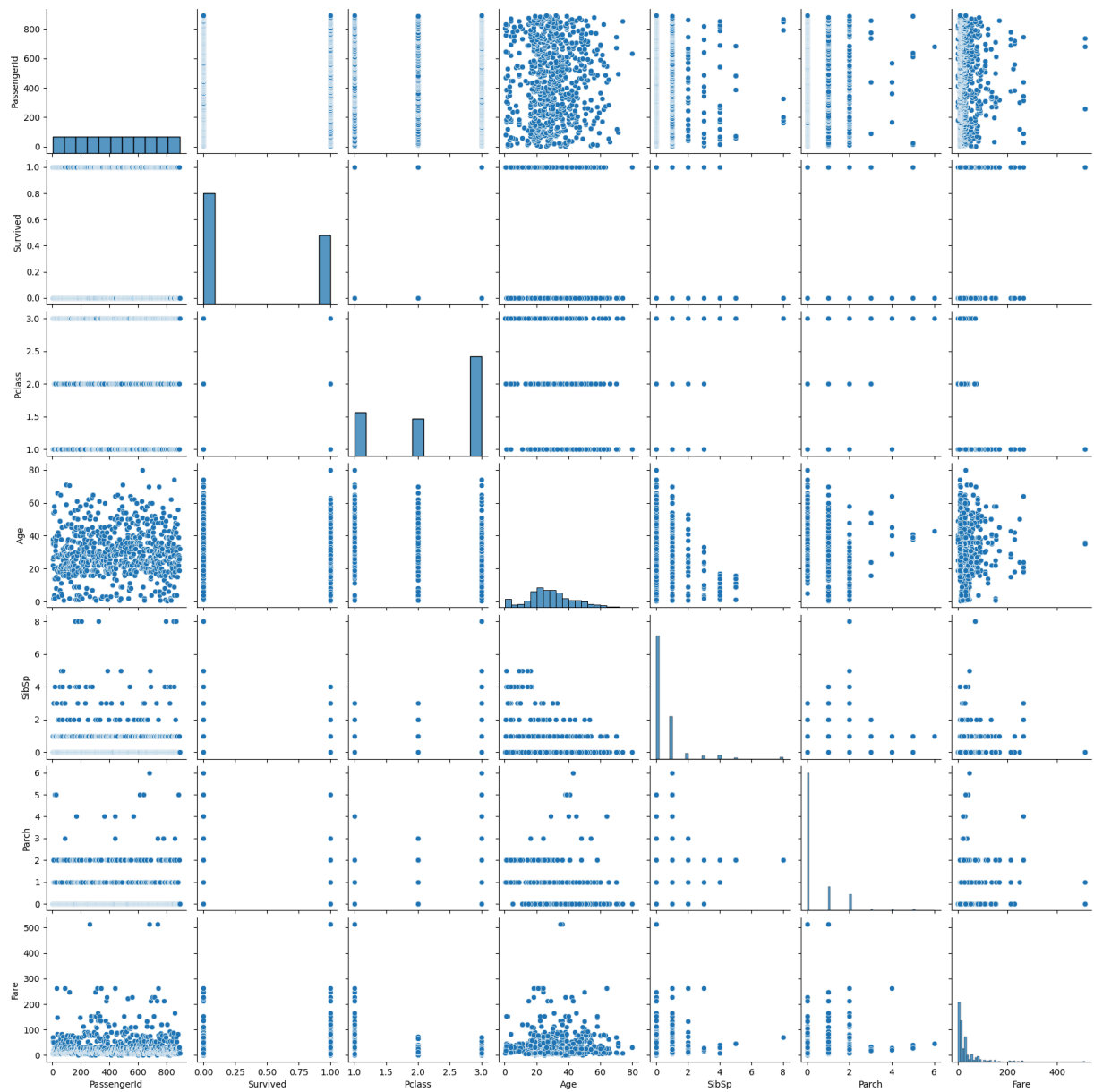
In []: The Fare column contains several outliers **with** unusually high ticket prices.

```
In [10]: plt.figure(figsize=(8,6))
sns.heatmap(df.corr(numeric_only=True),
            annot=True,
            cmap='coolwarm')
plt.show()
```



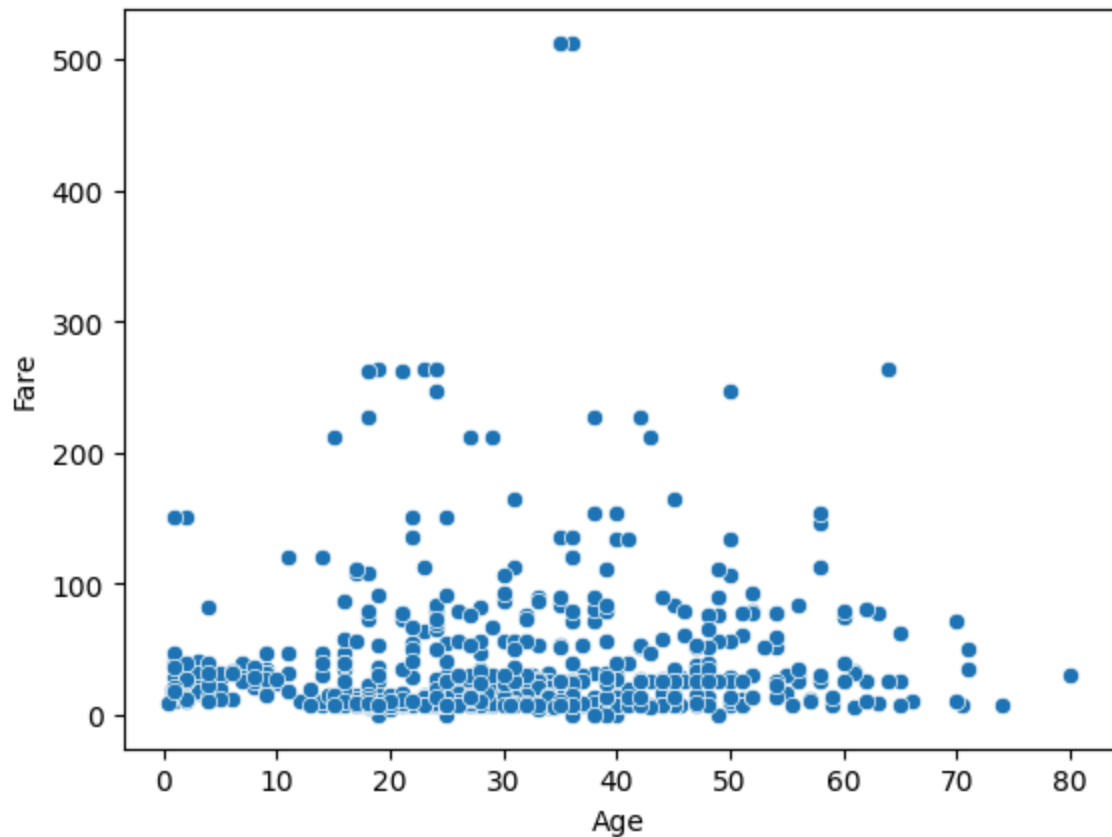
In []: The heatmap shows correlations between numerical variables. Survival has some relation with passenger class and fare

In [12]: `sns.pairplot(df)`
`plt.show()`



In []: The pairplot helps visualize relationships among Age, Fare, Pclass, and Survival.

```
In [15]: sns.scatterplot(x='Age', y='Fare', data=df)
plt.show()
```



In []: The scatter plot shows the relationship between Age and Fare. No strong relationship

In []: The Titanic dataset was analyzed using exploratory data analysis techniques. Missing Cabin, and Embarked columns. Histograms showed data distribution, boxplots revealed and heatmaps highlighted correlations. Pairplots and scatter plots helped identify Overall, EDA provided valuable insights into passenger characteristics and survival.